2 **(a)** obeys the equation pV = constant × T or $pV = nRT$	M1	
p, V and T explained	A1	
at all values of p, V and T/fixed mass/n is constant	A1	[3]
(b) **(i)** $3.4 \times 10^5 \times 2.5 \times 10^3 \times 10^{-6} = n \times 8.31 \times 300$	M1	
$n = 0.34$ mol	A0	[1]
(ii) for total mass/amount of gas		
$3.9 \times 10^5 \times (2.5 + 1.6) \times 10^3 \times 10^{-6} = (0.34 + 0.20) \times 8.31 \times T$	C1	
$T = 360$ K	A1	[2]
(c) when tap opened		
gas passed (from cylinder B) to cylinder A	B1	
work done <u>on</u> gas in cylinder A (and no heating)	M1	
so internal energy and hence temperature increase	A1	[3]